

Quantitative Trading: Theory and Practice

Advanced Course – Lecture 1

From Raw Data to Factor Testing

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Lecture 1 Positioning

- In the beginner part, we already introduced the basic ideas of factor investing, data cleaning, and portfolio construction.
- In this advanced lecture, we revisit factor research from a more practical and research-oriented perspective.
- The emphasis today is a complete workflow:

raw data $\rightarrow$ factor matrix $\rightarrow$ factor test $\rightarrow$ portfolio interpretation.

- The lecture is designed to be close to a real research script rather than only a conceptual overview.

Learning Objectives

By the end of this lecture, students should be able to:

- organize raw market data into a panel structure suitable for factor research;
- construct factor matrices and understand their time alignment;
- perform industry neutralization and listing-age filtering;
- evaluate factors by IC, RankIC, grouped returns, and long–short spread;
- understand turnover and simple transaction cost adjustment in factor evaluation;
- connect factor testing results to later stock selection and portfolio construction.

Panel Representation of Data

In our implementation, factor research is naturally organized as a panel indexed by time and asset.

Matrix view

- rows: trading dates,
- columns: assets,
- entries: prices, returns, factor values, or weights.

For example:

$$F \in \mathbb{R}^{T \times N}, \quad R \in \mathbb{R}^{T \times N},$$

where

- $F_{t,i}$ is the factor value of asset i at date t ,
- $R_{t,i}$ is the return of asset i at date t .

Key point: once the data is expressed in matrix form, cross-sectional factor testing becomes straightforward.

Raw Data and the Research Universe

In cross-sectional factor research, the starting point is a panel of asset-level observations over time.

- market variables: prices, returns, turnover, volume;
- asset identifiers and trading dates;
- metadata such as listing dates and industry classifications.

Two questions arise immediately:

- Which assets should be included in the universe?
- Which cross-sectional differences should be treated as part of the factor, and which should be removed?

Universe Preparation: Why Filtering Matters

A factor is always tested on a specific universe rather than on all available assets.

Typical filtering principles include:

- exclude assets with insufficient trading history;
- remove observations with missing factor values or missing future returns;
- keep the cross section comparable across dates.

Key point: factor evaluation depends not only on the factor formula, but also on the definition of the investable universe.

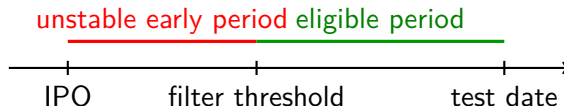
Filter 1: Recently Listed Stocks

Recently listed stocks are usually removed before factor testing.

Why filter them out?

- Their early prices may be dominated by IPO effects rather than normal trading behaviour.
- They often have limited historical data, so volatility, momentum, turnover, and other factors are less reliable.
- Their return distribution may be very different from mature stocks in the same cross section.

In practice, we may require that a stock has been listed for at least a fixed number of trading days before it enters the universe.



Filter 2: Missing Factor Values

A stock should be removed from the cross section on date t if its factor value is missing.

Why filter them out?

- IC, RankIC, and grouped returns all require a valid cross-sectional ranking on each date.
- Missing values may not be random; they can reflect suspension, data errors, or unavailable financial reports.
- Arbitrarily filling missing factor values may create artificial signals and distort the ranking.

A clean factor test should compare stocks with observable factor information:

$$\mathcal{U}_t^F = \{i \in \mathcal{U}_t : F_{t,i} \text{ is observed}\}.$$

Teaching point: missing-value handling is not only a technical step; it changes the actual research universe.

Filter 3: Suspended or Non-tradable Stocks

A factor test should approximate an implementable trading strategy.

Why filter them out?

- A suspended stock cannot be bought or sold at the rebalancing date.
- If a stock is not tradable, the theoretical factor portfolio may be impossible to implement.
- Including non-tradable stocks can overstate strategy performance and understate turnover risk.

A practical universe should therefore distinguish between

observable stocks and tradable stocks.

Important distinction

A stock may have a price and a factor value in the dataset, but it may still be unavailable for actual trading.

Filter 4: Special-treatment or Abnormal Stocks

In some markets, certain stocks are marked as special-treatment or abnormal-risk stocks.

Why filter them out?

- Their returns may be driven by regulatory status, distress risk, or restructuring events.
- They may face special trading rules, price limits, or liquidity constraints.
- A factor that works mainly on abnormal stocks may not represent a robust stock-selection signal.

For teaching and research, this filter helps students separate two different questions:

- Does the factor predict normal cross-sectional returns?
- Or is the result driven by a small group of abnormal-risk assets?

Filter 5: Low-liquidity Stocks

Low-liquidity stocks may look attractive in a backtest but be difficult to trade in practice.

Why filter them out?

- Their quoted prices may not represent executable prices for a realistic order size.
- Transaction costs and price impact are usually much higher.
- Small-cap or illiquid stocks may dominate backtest returns if no liquidity constraint is imposed.

Common liquidity filters include minimum trading volume, minimum turnover, or minimum market capitalization.

$$\mathcal{U}_t^{\text{liq}} = \{i : \text{turnover}_{t,i} \geq \tau_{\text{liq}}\}.$$

Key point: a factor is useful only if it can be converted into realistic trades.

Industry Neutralization

A factor may appear predictive simply because it loads on industry structure.

A common correction is industry neutralization:

- group assets by industry;
- standardize factor values within each industry;
- compare assets relative to peers in the same sector.

Within-industry standardization

For industry g ,

$$\tilde{F}_{t,i} = \frac{F_{t,i} - \mu_{t,g}}{\sigma_{t,g}}, \quad i \in g.$$

This helps separate stock selection information from broad sector effects.

Why Neutralization Is Needed

A raw factor may contain two different components:

- **stock-specific information**: useful for selecting stocks;
- **group-level exposure**: caused by industry, sector, size, or style structure.

Key idea

Neutralization removes unwanted group-level effects, so that assets are compared on a fairer basis.

For example, suppose a valuation factor is systematically lower in banks than in technology stocks. Then a strategy based on this factor may actually become:

“long banks, short technology”

rather than a true stock-selection strategy.

Example: A Misleading Factor Pattern

Suppose we test a valuation factor, such as book-to-price ratio.

In the raw cross section, we observe the following pattern:

High book-to-price stocks $\Rightarrow$ higher future returns.

At first glance, this seems to suggest a useful value factor.

However, after checking the portfolio composition, we may find:

High book-to-price portfolio $\approx$ mostly banks and real estate stocks,

while

Low book-to-price portfolio $\approx$ mostly technology and growth stocks.

Problem

The factor is not really comparing cheap stocks with expensive stocks. It is mainly comparing different industries.

Why This Pattern May Not Be Useful

The observed return difference may come from industry movement:

factor return $\approx$ return of banks and real estate – return of technology stocks.

This creates several problems:

- The strategy is making a hidden industry bet.
- The result may disappear when industry leadership changes.
- The factor may fail in a different market period.
- We cannot tell whether the signal comes from valuation or from sector exposure.

Key message

Without neutralization, a factor may look predictive only because it loads on industries that happened to perform well in the sample period.

What Neutralization Does in This Example

Industry neutralization changes the comparison.

Before neutralization:

Bank stocks vs. Technology stocks.

After neutralization:

cheap banks vs. expensive banks,

and

cheap technology stocks vs. expensive technology stocks.

Now the factor test asks a cleaner question:

Within the same industry, do cheaper stocks outperform more expensive stocks?

Interpretation

If the factor still works after neutralization, then it is more likely to contain genuine stock-selection information.

A Simple Decomposition View

Consider a raw factor value $F_{t,i}$ for stock i at time t .

If stock i belongs to industry g , we may decompose the factor as

$$F_{t,i} = \underbrace{\mu_{t,g}}_{\text{industry-level component}} + \underbrace{\varepsilon_{t,i}}_{\text{stock-specific component}}.$$

Here,

$$\mu_{t,g} = \frac{1}{N_{t,g}} \sum_{i \in g} F_{t,i}$$

is the average factor value within industry g .

Therefore, the stock-specific part can be estimated by subtracting the industry mean:

$$F_{t,i} - \mu_{t,g}.$$

Interpretation

After demeaning, a stock is evaluated relative to other stocks in the same industry.

From Demeaning to Standardization

Industry demeaning removes the industry average:

$$F_{t,i}^{\text{demeaned}} = F_{t,i} - \mu_{t,g}.$$

However, different industries may have different factor dispersions. For example, the same deviation of 0.2 may be large in one industry but small in another.

Therefore, we further divide by the within-industry standard deviation:

$$\sigma_{t,g} = \sqrt{\frac{1}{N_{t,g} - 1} \sum_{i \in g} (F_{t,i} - \mu_{t,g})^2}.$$

This gives the normalized factor:

$$\tilde{F}_{t,i} = \frac{F_{t,i} - \mu_{t,g}}{\sigma_{t,g}}, \quad i \in g.$$

Meaning: $\tilde{F}_{t,i}$ measures how many within-industry standard deviations stock i is above or below its industry average.

How the Neutralization Formula Is Derived

For each industry g , we want the transformed factor to satisfy two properties:

$$\frac{1}{N_{t,g}} \sum_{i \in g} \tilde{F}_{t,i} = 0, \quad \text{Std}_{i \in g}(\tilde{F}_{t,i}) = 1.$$

Starting from the raw factor $F_{t,i}$, subtract the within-industry mean:

$$F_{t,i} - \mu_{t,g}.$$

This guarantees that the transformed values have zero within-industry average.

Then divide by the within-industry standard deviation:

$$\tilde{F}_{t,i} = \frac{F_{t,i} - \mu_{t,g}}{\sigma_{t,g}}.$$

This makes the factor scale comparable across industries.

Result

The formula is simply a within-industry z-score transformation.

What Neutralization Changes in Factor Testing

Without neutralization, a factor test may mainly reflect industry allocation:

$$\text{factor return} \approx \text{industry return difference.}$$

After neutralization, the factor test focuses more on cross-sectional stock selection:

$$\text{factor return} \approx \text{within-industry stock selection effect.}$$

Industry-neutralized factors answer the question:

Is this stock attractive compared with similar stocks?

Key point

Neutralization does not make a factor “better” automatically. It makes the source of the factor signal cleaner and easier to interpret.

Limitations of Simple Industry Neutralization

Within-industry standardization is intuitive and easy to implement, but it has limitations.

- Small industries may have unstable estimates of $\mu_{t,g}$ and $\sigma_{t,g}$.
- Extreme outliers can distort the industry mean and standard deviation.
- It only removes industry effects, not other exposures such as size, beta, or liquidity.
- If the industry itself contains real predictive information, full neutralization may remove useful signal.

Practical workflow

A common pipeline is:

filtering $\rightarrow$ winsorization $\rightarrow$ standardization $\rightarrow$ industry neutralization $\rightarrow$ factor testing.

Neutralization should therefore be treated as a modeling choice, not a mechanical rule.

Factor Matrix and Time Alignment

After the universe is defined, the factor is organized into a matrix form:

$$F \in \mathbb{R}^{T \times N}, \quad R \in \mathbb{R}^{T \times N}.$$

- $F_{t,i}$: factor value of asset i at date t ;
- $R_{t,i}$: realized return of asset i at date t .

The crucial evaluation rule is:

$$F_t \longrightarrow R_{t+1}.$$

- the factor is measured at time t ;
- performance is evaluated using the next-period return;
- this is the most basic protection against look-ahead bias.

Two Core Diagnostics: IC and RankIC

The first family of factor evaluation methods studies cross-sectional association between factor values and future returns.

Normal IC

$$IC_t = \text{corr}(F_t, R_{t+1}).$$

RankIC

$$\text{RankIC}_t = \text{corr}_{\text{rank}}(F_t, R_{t+1}).$$

- IC focuses on linear association;
- RankIC focuses on ranking ability;
- both are computed cross-section by cross-section and then summarized over time.

Normal IC: What It Measures

Normal IC is the cross-sectional Pearson correlation between factor values and next-period returns:

$$IC_t = \text{corr}(F_t, R_{t+1}).$$

More explicitly, for a universe of N_t stocks at time t ,

$$IC_t = \frac{\sum_{i=1}^{N_t} (F_{t,i} - \bar{F}_t) (R_{t+1,i} - \bar{R}_{t+1})}{\sqrt{\sum_{i=1}^{N_t} (F_{t,i} - \bar{F}_t)^2} \sqrt{\sum_{i=1}^{N_t} (R_{t+1,i} - \bar{R}_{t+1})^2}}.$$

Interpretation

Normal IC asks whether stocks with higher factor values tend to have higher future returns in a linear sense.

Normal IC: Role, Advantages, and Limitations

Normal IC is commonly used to evaluate whether a factor has predictive power.

- If $IC_t > 0$, higher factor values are associated with higher future returns.
- If $IC_t < 0$, higher factor values are associated with lower future returns.
- If $IC_t \approx 0$, the factor has little linear predictive relation in that cross section.

Advantages

Normal IC is simple, intuitive, and directly measures linear cross-sectional association.

Limitations

Normal IC can be sensitive to outliers and may miss nonlinear or purely ordinal relationships.

RankIC: What It Measures

RankIC replaces raw factor values and raw returns by their cross-sectional ranks.

Let

$$\text{rank}(F_{t,i}) \quad \text{and} \quad \text{rank}(R_{t+1,i})$$

denote the ranks of stock i among all stocks in the universe.

Then RankIC is defined as

$$\text{RankIC}_t = \text{corr}(\text{rank}(F_t), \text{rank}(R_{t+1})).$$

Interpretation

RankIC asks whether the factor can correctly rank stocks by their future returns.

RankIC: Role, Advantages, and Limitations

RankIC is especially useful when we care more about stock ordering than exact factor magnitudes.

- A high RankIC means high-ranked factor stocks tend to become high-return stocks.
- It is suitable for portfolio construction based on top/bottom groups.
- It is less affected by extreme factor values or extreme returns.

Advantages

RankIC is more robust to outliers and monotonic transformations of the factor.

Limitations

RankIC ignores magnitude information. A small difference and a large difference in factor value are treated only through their ordering.

Normal IC vs. RankIC

Normal IC and RankIC answer related but different questions.

	Normal IC	RankIC
Input	Raw values	Ranks
Correlation type	Pearson correlation	Spearman correlation
Focus	Linear association	Ranking ability
Outlier sensitivity	Higher	Lower
Magnitude information	Preserved	Mostly ignored
Best use case	Linear factor-return relation	Long-short ranking strategy

Practical suggestion

In factor research, it is common to report both IC and RankIC. If both are stable and have the expected sign, the factor evidence is stronger.

How IC and RankIC Are Used Over Time

For each date t , we compute one cross-sectional IC or RankIC by aligning current factor values with next-period returns:

$$F_t \longrightarrow R_{t+1}.$$

Therefore, if the sample contains dates $1, 2, \dots, T$, the available IC series is:

$$IC_1, IC_2, \dots, IC_{T-1}.$$

The last factor cross section F_T cannot be used for IC calculation because R_{T+1} is not observed in the sample. Then we summarize the IC time series using statistics such as:

$$\overline{IC} = \frac{1}{T-1} \sum_{t=1}^{T-1} IC_t, \quad IR_{IC} = \frac{\overline{IC}}{\text{Std}(IC_1, \dots, IC_{T-1})}.$$

- $\overline{IC}$ measures the average predictive direction and strength.
- $\text{Std}(IC_1, \dots, IC_{T-1})$ measures the instability of the signal.
- IR_{IC} measures the stability-adjusted predictive power.

Valid Cross Sections and Missing Data

Cross-sectional factor tests should only compare assets with valid information.

- if factor values are missing, the asset cannot be ranked reliably;
- if future returns are missing, predictive relation cannot be evaluated;
- therefore, each date is tested on the intersection of valid assets.

Methodological principle

For each date, evaluation should be based on a common valid cross section of assets.

This avoids contaminating the test with artificial differences in data availability.

Grouped Return Analysis

Grouped return analysis evaluates whether assets with different factor levels earn systematically different future returns.

- sort assets by factor value at each date;
- divide them into Q groups, such as quintiles or deciles;
- compute the next-period average return of each group.

Grouped return series

$$R_t^{(1)}, R_t^{(2)}, \dots, R_t^{(Q)}.$$

This method is especially useful because it translates factor scores into an economically interpretable return structure.

Why Use Grouped Return Analysis?

IC and RankIC measure statistical association, but they do not directly show the economic return pattern.

Grouped return analysis answers a more practical question:

Do stocks with higher factor values actually earn higher future returns?

It is useful because it shows:

- whether the factor-return relation is monotonic;
- whether the signal is concentrated only in the top or bottom group;
- whether a long-short strategy is economically meaningful;
- whether the factor has stable portfolio-level implications.

Teaching point

IC tells us whether the factor is statistically related to future returns. Grouped returns show whether the factor can be translated into portfolio performance.

Step 1: Sort Stocks by Factor Values

At each date t , we start from the cross section of factor values:

$$F_{t,1}, F_{t,2}, \dots, F_{t,N_t}.$$

Then we sort stocks according to their factor values:

$$F_{t,(1)} \leq F_{t,(2)} \leq \dots \leq F_{t,(N_t)}.$$

Here, $(1), (2), \dots, (N_t)$ denote the sorted stock indices.

For example, if a larger factor value is expected to be better, then:

bottom group $\Rightarrow$ low factor stocks,

top group $\Rightarrow$ high factor stocks.

Key point

The grouping is always based on information available at time t , not on future returns.

Step 2: Divide Stocks into Groups

After sorting, we divide the universe into Q groups.

For example:

- $Q = 5$: quintile groups;
- $Q = 10$: decile groups.

Let $G_t^{(q)}$ denote the set of stocks in group q at time t :

$$G_t^{(q)} = \{i : \text{stock } i \text{ belongs to group } q \text{ according to } F_t\}.$$

Then the groups satisfy:

$$G_t^{(1)} \cup G_t^{(2)} \cup \dots \cup G_t^{(Q)} = \mathcal{U}_t,$$

where $\mathcal{U}_t$ is the tradable universe at date t .

Step 3: Compute Each Group's Next-Period Return

After forming groups at time t , we evaluate their next-period returns using R_{t+1} .

For an equal-weighted group portfolio, the return of group q is:

$$R_{t+1}^{(q)} = \frac{1}{|G_t^{(q)}|} \sum_{i \in G_t^{(q)}} R_{t+1,i}.$$

Here:

- $G_t^{(q)}$ is determined by factor values at time t ;
- $R_{t+1,i}$ is the next-period return of stock i ;
- $|G_t^{(q)}|$ is the number of stocks in group q .

Important alignment

The portfolio is formed at time t , but its performance is measured over the next period.

$$F_t \longrightarrow R_{t+1}^{(q)}.$$

Weighted Group Return

Sometimes we use weighted group returns instead of equal-weighted returns.

Let $w_{t,i}^{(q)}$ be the portfolio weight of stock i in group q . Then:

$$R_{t+1}^{(q)} = \sum_{i \in G_t^{(q)}} w_{t,i}^{(q)} R_{t+1,i},$$

where

$$\sum_{i \in G_t^{(q)}} w_{t,i}^{(q)} = 1.$$

Common choices include:

- equal weights;
- market-cap weights;

Equal-weighted group returns are easier to interpret, while weighted returns may be closer to implementable portfolio performance.

What Pattern Do We Expect?

If the factor is useful and larger factor values predict higher future returns, then we expect:

$$R_{t+1}^{(1)} < R_{t+1}^{(2)} < \dots < R_{t+1}^{(Q)}.$$

Over many dates, we usually compare average group returns:

$$\bar{R}^{(q)} = \frac{1}{T-1} \sum_{t=1}^{T-1} R_{t+1}^{(q)}.$$

A monotonic pattern across groups provides stronger evidence than only one good group.

Interpretation

A good factor should not only separate winners from losers; ideally, the return should improve gradually from low-score groups to high-score groups.

Long-Short Return from Group Analysis

Grouped return analysis naturally leads to a long-short portfolio.

If larger factor values are expected to be better, we may define:

$$R_{t+1}^{LS} = R_{t+1}^{(Q)} - R_{t+1}^{(1)}.$$

That is:

long top group + short bottom group.

The average long-short return is:

$$\bar{R}^{LS} = \frac{1}{T-1} \sum_{t=1}^{T-1} R_{t+1}^{LS}.$$

Economic meaning

The long-short return measures whether the factor spread can be turned into a portfolio return.

Grouped Return Analysis: Strengths and Limitations

Grouped return analysis is intuitive and economically interpretable.

Strengths

- directly shows portfolio-level return differences;
- helps detect monotonic factor-return relationships;
- connects factor testing with long-short strategy design;
- easier to explain than correlation-based diagnostics.

Limitations

- results may depend on the number of groups Q ;
- small groups may be noisy;
- transaction costs and turnover are not automatically included;
- group portfolios may still contain unintended industry or style exposures.

From Grouped Returns to Portfolio Interpretation

Grouped return analysis naturally leads to portfolio interpretation.

- **Monotonicity:** do higher factor groups earn higher future returns?
- **Long-only interpretation:** can the top group be viewed as a candidate buy portfolio?
- **Long-short interpretation:** does the spread between extreme groups generate a stable premium?

Extreme-group spread

If the factor is positively predictive,

$$LS_t = R_t^{(Q)} - R_t^{(1)}.$$

Thus grouped return analysis serves as the bridge between factor testing and portfolio thinking.

Long-Only and Long–Short Views of a Factor

A factor can be interpreted through at least two portfolio lenses.

Long-only view

- focus on the high-factor group;
- compare its return with the market average or the cross-sectional average.

Long–short view

- buy the high-factor group and short the low-factor group;
- study whether the factor spread delivers a persistent premium.

These two views answer different questions: whether the factor is useful for selection, and whether it generates relative-value alpha.

A factor may look strong statistically but still be unattractive in implementation.

- if the constituents of the high-factor or low-factor groups change too frequently, turnover will be high;
- high turnover leads to commissions, slippage, and weaker net performance;
- therefore, grouped portfolio analysis should be interpreted together with turnover.

Net factor spread idea

net spread return = gross spread return – trading cost adjustment.

This is one of the first places where statistical significance meets trading reality.

What a Complete Factor Evaluation Should Deliver

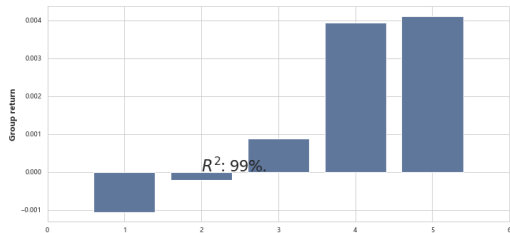
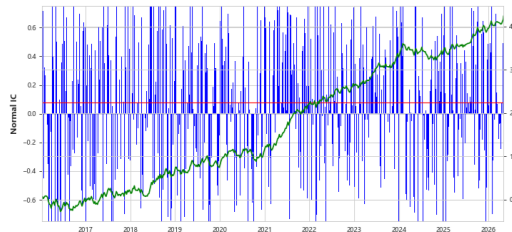
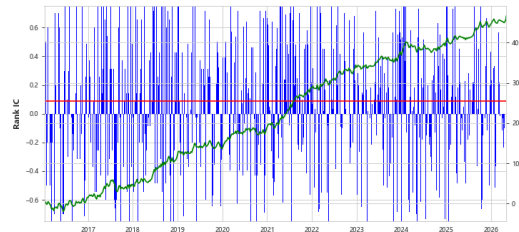
A complete factor evaluation should provide evidence from multiple angles.

- cross-sectional association: IC and RankIC;
- economic monotonicity: grouped return structure;
- portfolio interpretation: long-only and long-short cumulative performance;
- implementation awareness: turnover and simple cost sensitivity.

Key message: a factor should not be judged by a single statistic. It should be evaluated as both a predictive signal and a portfolio formation tool.

What a Complete Factor Evaluation Should Deliver

For example:



How to Read Grouped Return Results

When reading grouped return analysis, we should ask several questions.

- Is the return pattern approximately monotonic across groups?
- Does the top group outperform the bottom group by a meaningful margin?
- Is the high-factor group attractive from a long-only perspective?
- Is the long–short spread stable over time?
- Does the result remain attractive after considering turnover and simple costs?

Therefore: IC tells us whether the factor predicts returns, while grouped portfolio analysis tells us whether that predictive relation is economically useful.

From Factor Ranking to Stock Selection

Once a factor has passed initial evaluation, it can be used for stock selection.

- rank assets by factor value;
- select the strongest candidates;
- optionally retain part of the existing holdings to reduce unnecessary turnover.

Interpretation

Factor testing answers whether the factor ranks assets well. Stock selection uses that ranking to define a candidate portfolio universe.

This is the natural transition from factor research to portfolio construction.

From Portfolio Weights to NAV

After stock selection comes portfolio accounting.

- assign portfolio weights to selected assets;
- compute period-by-period profit and loss;
- track cumulative net asset value;
- compare strategy performance with a benchmark when needed.

Basic recursion

$$\text{NAV}_t = \text{NAV}_{t-1}(1 + \text{PnL}_t).$$

Although portfolio construction will be discussed more fully in the next lecture, the factor-testing stage already prepares the logic for this step.

Methodological Summary of Today

The methodology of this lecture can be summarized as follows:

- ① define the research universe from raw asset-level data;
- ② construct and preprocess the factor cross section;
- ③ align factor values with future returns correctly;
- ④ evaluate predictive ability using IC and RankIC;
- ⑤ evaluate economic usefulness using grouped returns and portfolio spreads;
- ⑥ examine implementation frictions such as turnover and simple trading costs.

In the programming session, students will implement a simplified factor research pipeline:

- ① prepare a panel of prices and returns;
- ② define a valid trading universe;
- ③ construct one factor and optionally neutralize it;
- ④ evaluate the factor by IC, RankIC, and grouped return analysis;
- ⑤ interpret the factor from both long-only and long-short perspectives;
- ⑥ discuss whether the factor remains attractive after considering turnover and simple costs.

- Factor research is a methodological pipeline from raw data to economic interpretation.
- A good factor evaluation should include both predictive diagnostics and grouped portfolio analysis.
- Long-only and long–short views provide complementary interpretations of the same factor.
- Turnover and simple trading costs are already relevant at the factor-testing stage.
- Next lecture: how to turn factor information into more systematic portfolio weights and strategy NAV.

Thank you!